BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 204 · Logic circuits, adders and flip-flops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why can a group of four 1s remove two variables?

Q2. Simplify ¬(A ∧ B).

Q3. Calculate 1+1 in binary.

Q4. What is (1 XOR 0) AND 1?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

half adder — Combinational circuit that adds two one-bit inputs.

full adder — Adds two bits and a carry-in bit.

Sum — Current-column result bit.

CarryOut — Bit passed to the next higher column.

D flip-flop — Clock-triggered circuit that stores one bit.

New learning

Find the Sum bit — XOR is 1 when A and B are different; A=1, B=1 gives Sum=0.

Find the Carry bit — AND is 1 only when both inputs are 1; here Carry=1.

Extend to full adder and memory — A full adder also includes CarryIn. Unlike combinational logic, a D flip-flop stores a bit until the triggering clock edge.

Worked example

A half adder receives A=0, B=1. Calculate Sum and Carry.

1. Predict and explain — XOR(0,1)=1, giving Sum=1.

2. Check and extend — AND(0,1)=0, giving Carry=0.

We Do · Practise with guidance

A full adder receives A=1, B=0, CarryIn=1. Calculate outputs.

$1 + 0 + 1 = 2 = 10_2$

Predict and explain

The total is decimal 2, binary 10.

Check and extend

Sum is the low bit 0; CarryOut is the high bit 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A half adder receives A=1, B=0. Give Sum and Carry.

Question 2

For a full adder A=1, B=1 and CarryIn=0, calculate Sum and CarryOut.

Question 3

A rising-edge D flip-flop has Q=0 and D=1. D changes to 0 before the next rising clock edge. What value is stored after that edge, and why?

Exit Ticket · Answer before checking

A D-type flip-flop always changes Q immediately when D changes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Two variables change within the group, so only constant variables remain.

Q2. ¬A ∨ ¬B.

Q3. 10_2.

Q4. 1. XOR gives 1, and 1 AND 1 is 1.

You Do pages · Independent questions

Q1. Sum=1; Carry=0.

Q2. $1+1+0=2=10_2$, so Sum=0, CarryOut=1.

Q3. Q remains 0 because it samples the current D=0 at the rising edge. Changes between edges do not update Q.

Exit Ticket · Model answer

False. An edge-triggered D flip-flop updates Q only at its triggering clock edge.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: